

LU9

A 64-bit hash finalizer measured in full, including the point at which it ends.

Origin, construction, mechanism, every measurement and every limit.

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The construction

```
uint64_t lu9(uint64_t x) {  
    x ^= 0x9E3779B97F4A7C15;  
    x = mulfold(x, 0x781F94B96E8EDB3B);  
    return mulfold(x, 0xB853D68343F7525B);  
}
```

mulfold(v, c) is the 128-bit fold: multiply v by c to the full 128 bits, then XOR the two 64-bit halves together. One line in C with `__uint128_t`, two instructions on x86-64 (mul + xor).

Property	Value
Cost (latency model of this work)	9 = 1 (constant) + 4 + 4 (two folds)
Class	hash finalizer — collisions allowed, not bijective
State	64 bit, no side effect, no memory
Avalanche (measured, 8192 inputs)	0.4999
Collisions	0 across 2^{20} consecutive inputs
Portability	anywhere $64 \times 64 \rightarrow 128$ -bit is available

The three constants do not come from this work alone: 0x9E37... is the classic Fibonacci constant, the two fold constants come from a search of our own. Composing them into this chain and measuring it is what this paper contributes; the shape is not (see Origin).

Origin: built against a measured weakness

lu9 came out of a defect. The starting point was the bare double fold — the same chain without the first line, cost 8:

```
x = mulfold(x, c1); return mulfold(x, c2);
```

It falls on counting inputs, on 11 of 32 disguises, and it does so with every one of the 500 constant combinations tested. The weakness sits in the structure. The choice of constant does not enter into it.

Where the weakness comes from.

Inputs of low multiplicative weight (small numbers, many zero bits at the bottom, which is to say exactly counters and IDs) produce a thin 128-bit product. The fold then has nothing to fold. The extreme case is visible to the naked eye: **the bare fold maps 0 to 0**, and does so with every constant. lu9, by contrast, returns 0x37F28927366A4E7B for 0.

Why no quick test finds this.

The bare K8 chain reaches **0.4990** in the avalanche measurement — practically the same as lu9 (0.4999) and as mix13 (0.5001). A test that averages over random inputs does not see a fixed point at zero, and sees a weakness on small numbers just as little; only the counter stream brings it to light.

The repair, and what it says about invention.

A single cheap operation up front makes every input generic before the first fold takes hold. Cost: +1. After that the chain passes all 32 entry streams. **Only after this construction did it become apparent that this is the building principle of the wyhash family** (Wang Yi): constant onto the input, then folds. That is an independent rediscovery, not an original design. The name lu9 denotes *this chain with these constants and this measurement*, not the shape.

The full measurement protocol

Measured on the RRC ladder: counting numbers in disguises (read backwards, complemented, rotated), judged by PractRand. Every stage with an anchor: a known weak mixer had to fall in the same run.

Stage	Scope	Result
Gauntlet	4 streams at 256 MB	holds, 4/4
Smoke	32 streams at 2 GB	holds, 32/32
Depth test	64 streams at 16 GB (2^{34})	holds, 64/64 — measured 30 Aug 2026
Diploma RRC-64-40	protocol 256 streams at 1 TB; 12 run, then stopped	FALLS — 10 held 1 TB, two fell at 2^{38} . The remaining 244 were never run: a candidate that has fallen twice does not need 244 more terabytes

The failure in detail.

Stream	Depth	Test	p-value
T0C0r40	2^{38}	[Low1/64] FPF-14+6/16	6.8×10^{-19}
T1C0r02	2^{38}	[Low8/64] FPF-14+6/16	3.0×10^{-27}

Two independent disguises, the same amount of data, the same test type, the same place in the output: the lowest bits. Alongside them PractRand reported over 900 further sub-tests with nothing to note.

The safeguards.

On exactly those two streams NASAM held the full 1 TB, so the measuring path judges correctly there. In the same run mix13 fell at 2^{19} , so the rig can reject. And the variant using addition instead of XOR fell as well, at 2^{39} .

The reach: what lu9 can do

The protocol yields a load-bearing statement of a kind that exists for hardly any finalizer:

On structured counter streams lu9 shows no detectable weakness up to at least 2^{37} bytes per disguise. From 2^{38} on, a weakness in the lowest bits is detectable.

For the usual purposes that is ample: a hash table, a sharding key or a fingerprint never sees 256 gigabytes of mixed output from a single counter sequence. Anyone building a generator for very long random streams, or running scientific simulations with extreme data volumes, is moving in the order of magnitude where this finding becomes relevant — and should take a mixer that has gone the distance. Since 28 August that no longer has to be the expensive option: meer10 (cost 10) passes RRC-64-40 on 256 streams of 1 TB, where lu9 tears at 2^{38} . NASAM (~15) and mx3 rev2 pass a stage deeper still (RRC-64-42) and remain the choice when that extra stage matters.

The relatives, measured

Chain	Cost	Avalanche	On the ladder
lu9 (^const + fold ×2)	9	0.4999	32/32, 2^{34} — falls at 2^{38}
mfa9 (+const instead of ^const)	9	0.5001	32/32 — falls at 2^{39}
bare fold ×2	8	0.4990	falls on 11/32, with 500 constants
wyrand mixing step ¹	5	—	falls at 2^{13} – 2^{14}
mix13 (Stafford)	12	0.5001	falls at 2^{16} – 2^{19}
meer10 (fold-xsh2-fold)	10	—	32/32, 2^{34} — passes the diploma: 256/256 at 2^{40}

¹ Original code from the wyhash source, verified bit-for-bit; measured as a finalizer on counter inputs, not as a generator (natively it runs on a counter with a large stride).

The avalanche column separates nothing. The cost-8 failure sits there next to lu9 and mix13, and all three look like model pupils.

What was NOT measured

What this paper does **not** answer:

Speed.

The cost figure 9 is a weight table, not a clock. As of 31 August 2026 one clock exists: a dependent chain on a Ryzen 7 5700X puts lu9 at **2.85 ns per call**, against 2.58 for the bare fold pair and 3.28 for meer10 (100 million iterations, three runs, `results/cost_bench.json`). One machine, one compiler, latency only. Throughput, code size and any other CPU are still unmeasured, and for a decision they have to be measured on the target.

Collision behaviour at scale.

2^{20} consecutive inputs were checked (zero collisions). A sample of that size says nothing about collision quality. As a fold, lu9 is not bijective; collisions necessarily exist.

Behaviour on real keys.

What was measured is structured counter streams — the hardest realistic case for a finalizer. How lu9 behaves on strings, on records, or inside a complete hash function is answered by other tools (SMHasher and relatives).

Security.

lu9 is **not** cryptographic. It has no one-way property: with the constants known, a preimage is cheap to find. (As a fold it is not a bijection either, so it cannot be inverted as a function, which helps nobody: finding *some* input that maps to a given output is what an attacker needs, and that is easy.) It offers no protection whatsoever against deliberately constructed inputs. Anyone who has to fend off hash flooding needs a keyed hash with a secret key.

And the limit of the statement itself.

That lu9 tears at 2^{38} is measured. That it has no weakness below that is *not proven* — it has not been demonstrated, which is a different thing. A sharper tool could find something earlier.

Measuring it yourself

The rig is documented openly (feeder, drivers, PractRand with build instructions). lu9 runs there under its name or as a chain:

```
feeder lu9 0 0 40
feeder chain 0 0 40 3 6 9e3779b97f4a7c15 \
8 781f94b96e8edb3b 8 b853d68343f7525b
```

Both forms deliver byte-identical streams; the verdict depends only on the byte stream, not on the machine.

Provenance of the building blocks

Shape (constant up front, then folds): wyhash family, Wang Yi. **Fold as a mixing step**: Vladimir Makarov (mum-hash), carried on in the wyhash line. **Fibonacci constant**: classic, among others in splitmix64. **Zero fixed point of the fold and the remedy**: Orson Peters (2024). **Why the lowest bits are weak**: Klimov and Shamir's T-functions (2004) — carries only travel upwards, output bit k depends only on input bits 0 to k. **Test scheme**: Pelle Evensen's RRC. **Test program**: PractRand, Chris Doty-Humphrey. **Plotting cost against RRC test depth**: Martin Leitner-Ankerl (2020). **Fold constants, the composition, the full measurement protocol out to diploma distance and the name**: this work.

In brief

What it is	A 64-bit hash finalizer made of one constant and two 128-bit folds, cost 9.
Where it comes from	Constructed against a measured weakness (the bare fold has a fixed point at zero and falls on counter inputs); only afterwards recognised as the wyhash shape.
What it holds	32/32 disguises at 2 GB, 64/64 at 16 GB, ten streams past 1 TB.
Where it ends	2^38 (256 GB) on two of 256 streams, in the lowest bits, safeguarded three ways.
What it is good for	Hash tables, sharding, fingerprints. Not for very long generator streams, not for security.

The name lu9 is not an abbreviation.
For L.

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As of: 28 August 2026. Measuring rig: *The Ladder*. Cost frame: *The Cost Limit*. The diploma run in detail: *Where K9 Ends*. Its successor, one cost point higher and over the full distance: *MEER10*.